

Fisher Information: $I_n(\theta) = -\mathbb{E}_\theta[\frac{\partial^2}{\partial \theta^2} \ln(f(x|\theta))]$

Delta Method: $\sqrt{n}(g(\bar{x}) - g(\theta)) \xrightarrow{D} N(0, \sigma^2[g'(\theta)]^2)$

Score: $\psi(x; \theta) = \frac{\partial}{\partial \theta} \ln(f(x|\theta))$

Slutsky's Theorem:
$$\begin{cases} X_n + Y_n \xrightarrow{D} X + a \\ X_n Y_n \xrightarrow{D} aX \\ \frac{X_n}{Y_n} \xrightarrow{D} SOMETHING \end{cases}$$

CRLB: $\frac{(\tau'(\theta))^2}{I_n(\theta)}$

Regularity Conditions:

- Data are iid, parameter is identifiable, parameter is not used as a boundary of support

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2 Asymptotic Evaluation of Estimators

2.1 Consistency

Definition: Let W_n be an estimator based on sample of size n of $\tau(\theta)$. W_n is said to be consistent for $\tau(\theta)$ if $W_n \xrightarrow{P_\theta} \tau(\theta) \forall \theta$

- ie. Bias goes to 0, variance goes to 0

- $P_\theta(|W_n - \tau(\theta)| > \epsilon) \xrightarrow{n \rightarrow \infty} 0, \forall \theta$ for any $\epsilon > 0$

Methods:

1. Tail/Concentration Bounds to control the probabilities: Markov's Inequality, Chebychev, Chernoff

2. If $W_n = g(\bar{x})$, we can apply LLN ($\bar{x} \xrightarrow{P} \mu$) then CMT (Continuous Mapping Theorem) ($g(\bar{x}) \xrightarrow{P} g(\mu)$) if g is continuous.

3. M-estimation (Maximizer/Minimizer of some function, ie. MLE): $\frac{1}{n} l(\theta|x) \xrightarrow{P_{\theta_0}} E_{\theta_0}[\ln(f(x_1|\theta))]$

- Basically, we know the right side is min/max at θ_0 , so we need to show that the left side approaches that same min/max as the overall distributions converge

2.2 Asymptotic Variance

Definition: If $\lim_{n \rightarrow \infty} b_n \text{Var}(W_n) \rightarrow \gamma^2$, then γ^2 is called the limiting variance of W_n

Definition: If $\sqrt{n}(W_n - \tau(\theta)) \xrightarrow{D} Y_\theta, \forall \theta$, where $\mathbb{E}_\theta[Y_\theta] = 0$, then $\text{Var}_\theta(Y_\theta)$ is called the asymptotic variance of W_n .

- $\text{Var}_\theta(W_n) = \frac{1}{n} \text{Var}_\theta(\sqrt{n}(W_n - \tau(\theta))) \approx \frac{\text{Var}_\theta(Y_\theta)}{n}$

- "1-2" punch for MVUEs: Law of Large numbers, continuous mapping theorem

- "1-2" punch for Asymptotics: Central Limit Theorem, Delta Method

Define: An estimator W_n is said to be:

- i) Asymptotically unbiased if $\sqrt{n}(W_n - \tau(\theta)) \xrightarrow{D} Y_\theta, \mathbb{E}_\theta[Y_\theta] = 0$
- ii) Asymptotically efficient if its asymptotic variance is $\frac{[\tau'(\theta)]^2}{I_1(\theta)}$ (the CRLB, Fischer information calculated for 1 sample)

Theorem: Under regularity conditions, if $X \stackrel{\text{iid}}{\sim} f(x|\theta)$, then:

- i) the MLE ($\hat{\theta}_n$) will exist and be unique with probability converging to 1
- ii) $\sqrt{n}(\hat{\theta}_n - \theta_0) \xrightarrow{D} N(0, I_1^{-1}(\theta_0))$

2.3 Relative Efficiency

Suppose W_n and V_n are both asymptotically unbiased and have asymptotic variances σ_W^2, σ_V^2

Definition: The relative asymptotic efficiency of V_n compared to W_n is $ARE(V_n, W_n) = \frac{\sigma_W^2}{\sigma_V^2}$ (an estimator is better when its asymptotic variance is smaller, so this says that relative efficiency increases as the target estimator's variance decreases compared to another's).

3 Hypothesis Testing

Definition: Let $\mathbb{X}$ be the sample space. A test function is a statistic $\phi(x) \in [0, 1]$ such that if $x \in \mathbb{X}$, we reject H_0 with probability $\phi(x)$.

Definition: A simple test function satisfies $\phi(x) \in \{0, 1\} \forall x \in \mathbb{X}$

Rejection Region: $R_\phi = \{x \in \mathbb{X} : \phi(x) = 1\}$

Acceptance Region: $A_\phi = \{x \in \mathbb{X} : \phi(x) = 0\}$

Type I Error: $P_\theta(\text{Reject } H_0), \theta \in \Theta_0$ (Reject when we shouldn't)

Type II Error: $1 - P_\theta(\text{Reject } H_0), \theta \in \Theta_1$ (Accept when we shouldn't)

Power: $P_\theta(\text{Reject } H_0) = \mathbb{E}[\phi(X)] =: \beta_\phi(\theta)$

- Under Null, that's Type I Error Rate

- Under Alternate, that's just the power

Size: The size of ϕ is $\sup_{\theta \in \Theta_0} \beta_\phi(\theta)$

Level: ϕ is a level α test if its size is no more than α

3.1 Most Powerful Tests

Define: A test $\phi^* \in \Phi_\alpha$ is said to be MP (Simple H_0) or UMP (Composite H_0) for $H_0 : \theta \in \Theta_0$ vs. $H_1 : \theta \in \Theta_1$ if $\beta_{\phi^*}(\theta) \geq \beta_\phi(\theta) \forall \phi \in \Phi_\alpha$ and all $\theta \in \Theta_1$.

3.2 Neyman-Pearson Lemma

Consider testing $H_0 : \theta = \theta_0$ vs. $H_1 : \theta = \theta_1$ (both are simple). Then any test of the form (with $k > 0, \gamma \in [0, 1]$):

$$\phi(x) = \begin{cases} 1, & f(x|\theta_1) > kf(x|\theta_0) \\ \gamma, & f(x|\theta_1) = kf(x|\theta_0) \\ 0, & f(x|\theta_1) < kf(x|\theta_0) \end{cases}$$

is an MP test of a level equal to its size.

3.3 MLR

Let T be any univariate statistic and $g(t|\theta)$ be its PDF/PMF. We say $g(t|\theta)$ has the monotone likelihood ratio (MLR) property if: $\frac{g(t|\theta_2)}{g(t|\theta_1)}$ is non-decreasing in t whenever $\theta_2 > \theta_1$.

3.4 Karlin-Rubin

Let $f(x|\theta)$ be a family with univariate sufficient statistic T that has MLR. Then any test function of the form $\phi(x) = \begin{cases} 1, & T(X) > t_0 \\ \gamma, & T(X) = t_0 \\ 0, & T(X) < t_0 \end{cases}$ is the UMP test of level $\gamma = P_{\theta_0}(T > t_0)$ for $H_0 : \theta \leq \theta_0, H_1 : \theta > \theta_0$.

H_0/MLR	Non-Dec	Non-Inc
$\theta \leq \theta_0$	$T > t_0$	$T < t_0$
$\theta \geq \theta_0$	$T < t_0$	$T > t_0$

3.5 Likelihood Ratio Tests

Definte: The (generalized) Likelihood ratio is $\lambda(x) = \frac{\sup_{\theta \in \Theta_0} L(\theta|x)}{\sup_{\theta \in \Theta} L(\theta|x)}$

Define $\hat{\theta}$ to be the ordinary MLE and $\hat{\theta}_0 = \underset{\theta \in \Theta_0}{\operatorname{argmax}} L(\theta|x)$. Then $\lambda(x) = \frac{L(\hat{\theta}_0|x)}{L(\hat{\theta}|x)}$

We reject when $\lambda(x)$ is small (ie. the likelihood of the estimator being in the null space is small compared to main).

- $P_{\theta_0}(T(X) \leq k_1) + P_{\theta_0}(T(X) \leq k_2) = \alpha, \lambda(k_1) = \lambda(k_2)$

3.6 Wilks' Theorem

Under regularity assumptions, if $H_0 : \theta = \theta_0$ is true, $\Theta \in \mathbb{R}^k : -2\ln(\lambda(x)) \xrightarrow{D} \chi_k^2$

- Part 2: $X_i \stackrel{\text{iid}}{\sim} f(x|\theta), H_0 : \theta \in \Theta_0, H_1 : \theta \notin \Theta_0, \Theta_0 = \{\theta \in \mathbb{R}^k : \theta_j = \theta_{0,j}, j = 1, \dots, r\}$. Then, under $H_0 : -2\ln(\lambda(x)) \xrightarrow{D} \chi_r^2$

3.7 Wald Tests

Wald Statistic: $Z_w = (\hat{\theta} - \theta_0) \sqrt{nI_1(\hat{\theta})} \sim N(0, 1)$

Wald Test: Rejects when $|Z_w| > Z_{1-\alpha/2}$

3.8 Score Tests

Remember: $\sqrt{n}\psi(x; \theta_0) \xrightarrow{D} N(0, I_1(\theta_0))$

Score Statistic: $Z_s = \frac{\psi(x; \theta_0)}{\sqrt{nI_1(\theta_0)}}$

Score Test: Rejects when $|Z_s| > Z_{1-\alpha/2}$

3.9 Hypothesis Testing Summary

1) Definitions:

2) UMP:

- Neyman-Pearson (simple H_0, H_1)
- Karlin-Rubin (one-sided w/ MLR)
- Beyond that, no guarantee for UMP

3) LRT

- Reframe the test: $\phi(x) = I(\lambda(x) < c)$ as $\phi(x) = I(T(X) < b_1) + I(T(X) > b_2)$

4) Asymptotic Tests

- Wilks' theorem
- Wald/Score tests

4 Confidence Intervals

Coverage Probability: The Coverage Probability of an interval estimate $(L(X), U(X))$ at $\theta \in \Theta$ is $P_\theta(L(X) \leq \theta \leq U(X))$

Confidence Coefficient: The Confidence Coefficient of an interval estimate $(L(X), U(X))$ is $\inf_{\theta} P_\theta(L(X) \leq \theta \leq U(X))$ (the min).

- Similar to Level vs. Size, an interval estimate is level $100(1 - \alpha)\%$ if its confidence coefficient is at least $1 - \alpha$.

4.1 Pivots

Definition: A random variable $Q(x, \theta)$ is said to be pivotal if its distribution does not depend on θ

Location: $f(x|\theta) = g(x - \theta)$

- Pivots: $X_i - \theta$, and any function of that value (ie. $\bar{X} - \theta$, $\tilde{X} - \theta$ (median), any $h(X_1 - \theta, \dots, X_n - \theta)$)

Scale: $f(x|\theta) = \frac{1}{\theta} g(\frac{x}{\theta})$

- Pivots: $\frac{X_i}{\theta}$, and any function of that (ie. $\bar{X}/\theta$, $\frac{\tilde{X}}{\theta}$, and $h(\frac{X_1}{\theta}, \dots, \frac{X_n}{\theta})$)

If both, do the location centering first, then the scale: $\frac{X_1 - \theta}{\sigma}$

Confidence Set: A confidence set $C(x) \subset \mathbb{R}$ has level $1 - \alpha$ if $P(\theta \in C(x)) \geq 1 - \alpha$

- If $Q(X, \theta)$ is a pivot, then $C(x) = \{\theta : a \leq Q(X, \theta) \leq b\}; a < b$ forms a confidence set

- Let D be CDF of Q . Then $P_\theta(\theta \in C(x)) = P_\theta(a \leq Q(X, \theta) \leq b) = D(b) - D(a)$. To ensure $C(x)$ has level $1 - \alpha$, choose α_1, α_2 such that $\alpha_1 + \alpha_2 = \alpha; a = D^{-1}(\alpha_1), b = D^{-1}(1 - \alpha_2)$

- Typically, $Q(X, \theta)$ is decreasing in θ , so for $a \leq Q(X) \leq b$ there are corresponding values for $\theta : L(X), U(X)$ (think of the graphs of a decreasing function, where a maps to $U(X)$, etc.)

Uniform Transformation: Remember that if T is some statistic with continuous CDF $F_T(t|\theta)$, then $U = F_T(t|\theta) \sim U(0, 1)$ (ie. $F^{-1}(U) = T$)

- Choose $Q(X, \theta) = F_T(T|\theta)$ as a pivot: $C(x) = \{\theta : a \leq Q(X, \theta) \leq b\}$ is a confidence set with coefficient $b-a$ for $0 \leq a < b \leq 1$

4.2 Inverting Tests

Given a confidence set $C(x)$ for $H_0 : \theta = \theta_0$, with test function $\phi(x) = \begin{cases} 1, & \theta_0 \notin C(x) \\ 0, & \theta_0 \in C(x) \end{cases}$, the size is $P_{\theta_0}(\theta_0 \notin C(x)) = 1 - P_{\theta_0}(\theta_0 \in C(x))$

We can operate in reverse order:

Theorem: For each $\theta_0 \in \theta$, let $A(\theta_0)$ be the acceptance region of a size α test of $H_0 : \theta = \theta_0, H_1 : \theta \neq \theta_0$

- Then, $C(x) = \{\theta_0 : x \in A(\theta_0)\}$ is a level $100(1 - \alpha)\%$ confidence set

4.3 Length Minimization

Assuming we have some confidence interval $(a \leq F(\theta) \leq b)$, subject to constraint $F(b) - F(a) = 1 - \alpha$ (ie. size α).

Lagrangian: $h(a, b, \lambda) = b - a + \lambda[F(b) - F(a) - (1 - \alpha)]$

- It's composed of: 1) The part we want to minimize (the interval length) and b) λ times our constraint (written as being equal to 0)

Then take partials of everything:

- $\frac{\partial}{\partial a}$ (solve for λ , ie. $\lambda = \frac{1}{f(a)}$)

- $\frac{\partial}{\partial b}$ (solve for λ , ie. $\lambda = \frac{1}{f(b)}$, and set the first two parts equal to each other (ie. $\frac{1}{f(a)} = \frac{1}{f(b)}$))

- $\frac{\partial}{\partial \lambda} = F(b) - F(a) - (1 - \alpha)$ (returns our constraint)

That gives 2 equations, 2 unknowns. Can use various root finding methods to solve.

4.4 Large Sample Intervals

When n is large it can be beneficial to use 1) Asymptotic Pivots or 2) Invert large sample tests

For $H_0 : \theta = \theta_0, H_1 : \theta \neq \theta_0$:

Wald Intervals

- $Z_w = \frac{\hat{\theta} - \theta_0}{\text{Std.Error}(\hat{\theta})} \xrightarrow{H_0} N(0, 1)$

- $A(\theta_0) = \{x : |Z_w| \leq Z_{1-\alpha/2}\}$

- $C(x) = \{\theta_0 : |Z_w| \leq Z_{1-\alpha/2}\} = [\hat{\theta} - Z_{1-\alpha/2}SE(\hat{\theta}), \hat{\theta} + Z_{1-\alpha/2}SE(\hat{\theta})]$

- This is asymptotic, so as n gets larger, test approaches the $N(0, 1)$ and approaches the size α test

- $Q(X, \theta) = \frac{\hat{\theta} - \theta}{SE(\hat{\theta})} \xrightarrow[\text{large } n]{} N(0, 1)$ leads to large sample pivotal interval: $[\hat{\theta} - SE(\hat{\theta})Z_{1-\alpha/2}, \hat{\theta} + SE(\hat{\theta})Z_{1-\alpha/2}]$

Score Intervals

- $Z_s = \frac{\psi(x; \theta)}{\sqrt{nI_1(\theta_0)}} \xrightarrow{H_0} N(0, 1)$

- Using this as an asymptotic pivot or inverting an asymptotic size α test (the score test) gives: $C(x) = \{\theta_0 : |\frac{\psi(x;\theta_0)}{\sqrt{nI_1(\theta_0)}}| \leq Z_{1-\alpha/2}\}$

LR Intervals

- $-2\ln(\lambda(x)) \xrightarrow[H_0]{D} \chi_1^2$ (for one parameter)
- Test-based confidence set: $C(x) = \{\theta_0 : -2\ln(\lambda(x)) \leq \chi_{1-\alpha}^2\}$ (θ_0 is used in the numerator of $\lambda(x)$)
- Pivot-based confidence set: $C(x) = \{\theta_0 : \chi_{\alpha, \alpha_1}^2 \leq -2\ln(\lambda(x)) \leq \chi_{1, 1-\alpha_2}^2\}$

Transformation-based Interval

$$\hat{\sigma}_{\hat{\theta}} = \sqrt{\frac{1}{sI_1(\hat{\theta})}}$$

- Start with $\sqrt{n}(W_n - \tau(\theta)) \xrightarrow{D} N(0, \nu_{\theta}^2)$
- Want $g'(\theta) = \frac{1}{\nu_{\theta}}$, then by Delta Method:
- $Q(X, \theta) = \sqrt{n}(g(W_n) - g(\theta)) \xrightarrow{D} N(0, 1)$